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CS313 - DATA MINING

Final Project Report

PREDICTING CITATION TREND IN COMPUTER SCIENCE PUBLICATIONS

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1 Introduction

1.1 Problem Statement

In higher education and scientific research, the assessment of research quality often relies on lagging indicators such as citation counts. The explosive growth of scientific publications in the field of Computer Science (CS), particularly in Artificial Intelligence (AI) and Machine Learning (ML) since 2015, has created a massive volume of data that poses significant challenges for screening and evaluation. In reality, citation counts are often slow to react, typically peaking several years after a paper is published. This makes it difficult for universities and research institutes to assess the early potential of research works, allocate resources effectively, and guide research topics for learners.

1.2 Objectives

This study aims to develop a method for predicting the popularity of scientific papers in Computer Science—specifically the number of citations in subsequent years—to achieve the following specific objectives:

- **Develop a Dataset and Forecasting Model:** Collect and integrate data from reputable sources such as **arXiv**, **Google Scholar**, **Semantic Scholar**, **Hugging Face**, **CORE**, and **Scimago** to build a prediction model based on relevant features (content, author, venue, citation, etc.).
- **Evaluate Research Potential:** Identify works with the potential for high impact.
- **Support Academic Orientation:** Assist students and lecturers in selecting influential research topics, particularly in rapidly evolving fields such as Computer Vision (cs.CV), Machine Learning (cs.LG), and Artificial Intelligence (cs.AI).
- **Policy Planning and Collaboration:** Provide an objective database and forecasting information to support administrators in planning research policies, allocating budgets, and fostering effective scientific collaboration networks.



2 Web Crawling and Data Collection

To construct a dataset for predicting citation trends in Computer Science (CS) research publications, we collected data from multiple reputable sources to ensure completeness, reliability, and accessibility of open data. The selected sources cover paper content, author information, citation metrics, and other metadata such as papers' venue, authors, etc., with a primary focus on publications from 2012 to 2024.

- **arXiv:** serves as the primary data source. We collect metadata on key fields such as `arxiv_id`, `title`, `abstract`, `authors`, `categories`, `published_date`, and `num_revisions`.
- **Google Scholar and Semantic Scholar:** were used to enrich citation-related information and venue rankings as well as authors' metadata. We crawled `citationCount`, `citations_by_year`, `venue_name` and authors' `citations`, `h-index`, `i10-index`, `influential_citation_count`.
- **Hugging Face:** was leveraged to obtain application-level indicators such as `github_stars`, `citing_models`, and `upvote counts` and `citing_datasets`.
- **CORE and Scimago:** were used to supplied venue ranking information (Q1-Q4/A*-C).

The crawling process was fully automated using Python, with the `requests`, `selenium`, and `beautifulsoup4` libraries. After data cleaning and filtering, the final dataset consists of 15,959 papers, stored in JSON format.

3 Exploratory Data Analysis and Data Preprocessing

3.1 Exploratory Data Analysis

The raw dataset contains 23 features, which are organized into three main groups based on their semantic characteristics and their intended roles in the predictive modeling pipeline. Table 1 provides a detailed description of each feature. The temporal evolution of publications and total citations across Computer Science subfields from 2012 to 2024 is illustrated in Figure 1. Figure 2 presents the total number of citations aggregated by primary research category (top 15 categories). Citation distributions across normalized venue rankings are shown in Figure 3 on a logarithmic scale.

3.2 Data Preprocessing and Feature Engineering

The detailed process of data preprocessing is shown in Table 2.

Feature Engineering:

- For papers with abnormally high citation counts in the "Others" (see Figure 3) category, we reassign each paper to the venue ranking or venue type whose median citation count is closest to that paper's citation value.
- For each category, we extract two temporal trend features to capture the growing trend of categories over years: `slope_category_papers`, `slope_category_citations`. These two features can be used to model the slopes of both annual publication counts and citation counts over time.



Group	Features	Description
Topic-related (Content & Topics)	title, abstract, categories, primary_category, keywords	These fields contain textual content and topical classifications. They are used to extract semantic representations (e.g., text embeddings) and to characterize the “hotness” or popularity of research topics over time.
Metadata (Publication Metadata)	authors, published_date, num_revisions, num_pages, citations_by_year, citationCount, citations, referenceCount, references, influentialCitationCount, venue_type, venue_ranking	These features provide publication context. Among them, citationCount is defined as the target variable to be predicted. Features such as authors and venue_ranking are expected to be strong predictors of citation impact.
Popularity (Community Engagement)	github_stars, upvote, citing_models, citing_datasets, citing_spaces, citing_collections	These indicators measure community interest and adoption across platforms such as GitHub and Hugging Face. The coverage and completeness of these features are carefully evaluated before inclusion in the final model.

Table 1: Feature Description

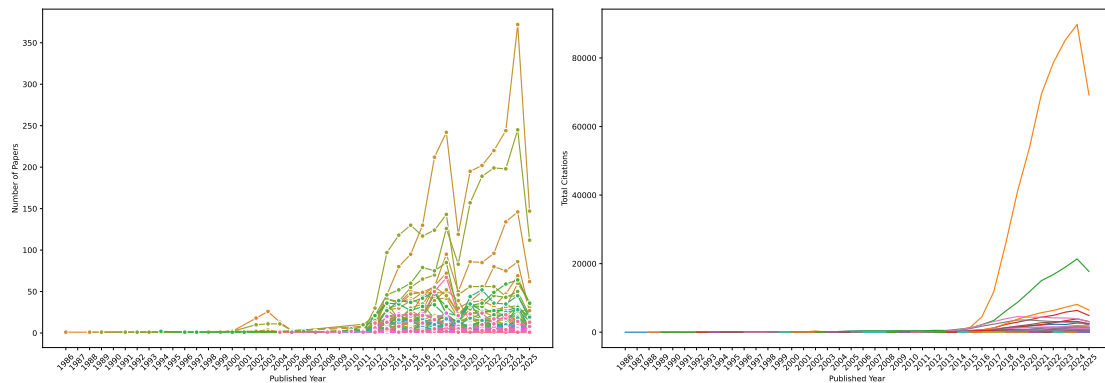


Figure 1: Temporal trends of publications and citations by primary research category (2012-2024). A sharp increase in publication volume is observed after 2015, with a pronounced peak during 2020-2021. Citation counts exhibit a delayed response, continuing to rise and reaching their maximum several years after the publication peak, highlighting the time-lagged nature of academic impact.

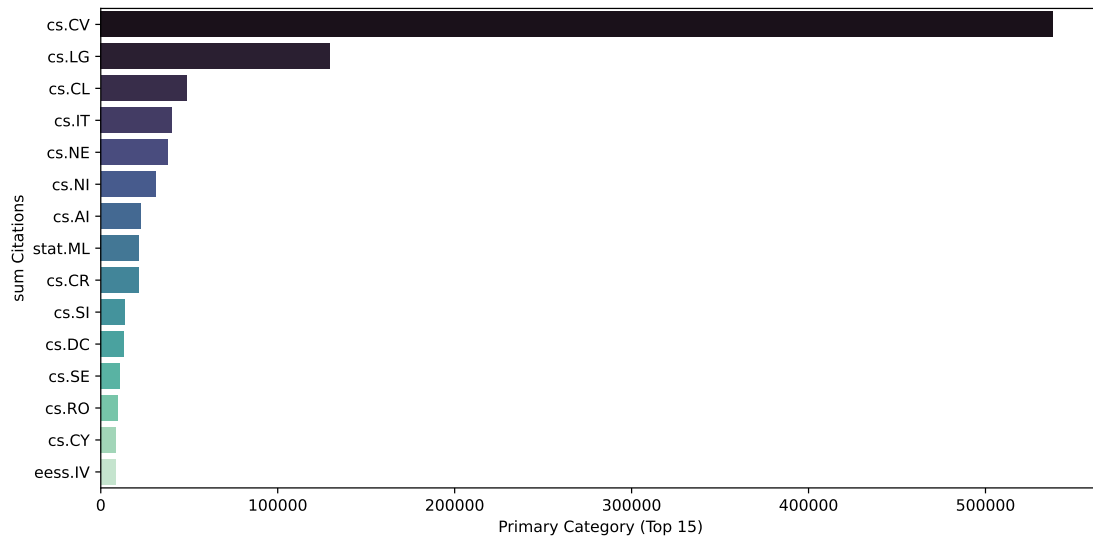


Figure 2: Total number of citations aggregated by primary research category (top 15 categories) (2012-2024). Vision- and learning-related fields such as cs.CV and cs.LG dominate citation volume, reflecting their high research activity and impact. Other categories show substantially lower citation totals, indicating more specialized or smaller research communities.

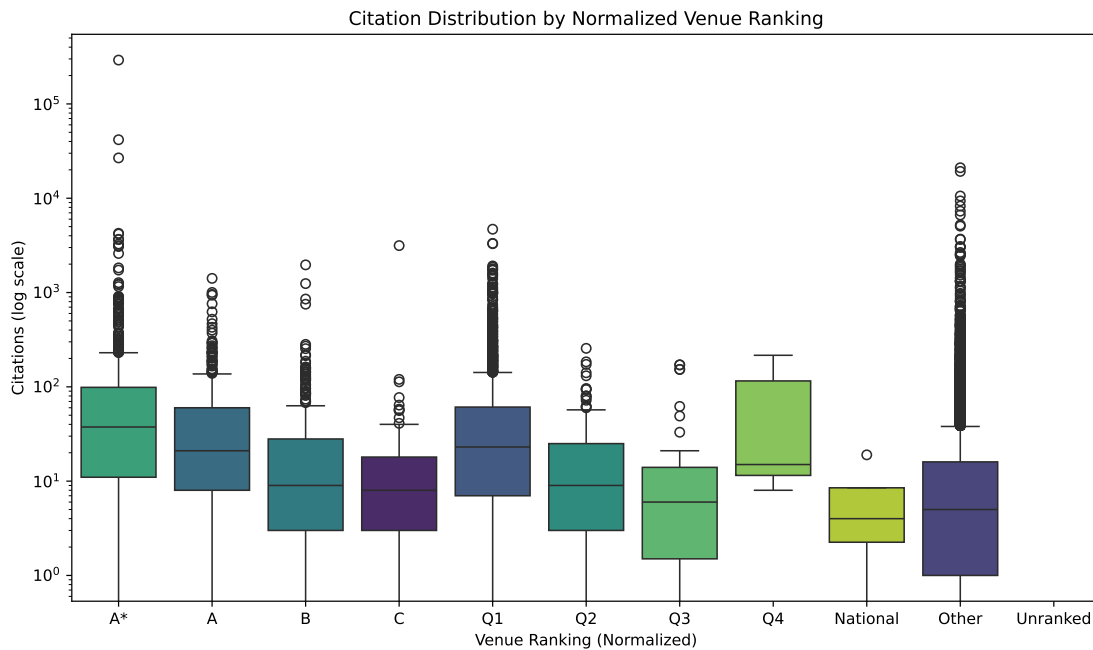


Figure 3: Citation distributions across normalized venue rankings (2012-2024). Higher-ranked venues (A*, A, Q1) generally exhibit higher median citation counts and greater variability, reflecting stronger academic impact. Lower-ranked and unranked venues tend to have lower medians but still include notable high-citation outliers. This suggests a mismatch between venue rankings and paper citation counts, likely arising from outdated venue information on some platforms.



Feature	Issue	Processing Action	Rationale
venue	Nested structure	Split into venue_name, venue_type, venue_ranking	Normalize publication venue information
venue_name, pdf_url, embedding	Irrelevant for modeling	Drop columns	Not required for training
num_pages	Missing values	Drop rows	Insufficient basic metadata
github_stars	Missing values	Fill with 0	Assume no public GitHub popularity
venue_ranking	Missing values, categorical data	Fill with 0 & Label encoding	Represent unranked venues
venue_type	Missing values, categorical data	Fill with "preprint" & Label encoding	Reflect arXiv preprint publications
categories, primary_category	Inconsistent formatting, categorical data	Regex extraction (cs.XX) & Label encoding	Standardize research field labels
title, abstract	Noisy text	Lowercasing and symbol removal	Prepare clean textual data
citations_by_year	Temporal structure	Expand into citations_YYYY	Convert to tabular format
citations before publication year	Temporally invalid	Assign -1	Explicitly mark invalid time entries
arxiv_id	Duplicate records	Drop duplicates	Ensure one record per paper

Table 2: Data Preprocessing



- **Text-based features (title, abstract):** We employ BERTopic to discover latent research topics from paper titles and abstracts, and leverage topic-related information as input to our downstream citation prediction model. Specifically, each paper's topic is encoded using one-hot representations, complemented by temporal topic features—`slope_topic_papers` and `slope_topic_citations`—which capture the evolution of topic popularity and impact over time.

3.3 Text and Temporal Data Mining

We employ specific data mining techniques to transform unstructured text and temporal data into structured features for our predictive models.

Text Mining with BERTopic: To capture the semantic content of research papers, we apply BERTopic to the `title` and `abstract` fields. Documents are embedded using the `allennai-specter` model, reduced in dimensionality via UMAP, and clustered using HDBSCAN to identify latent topics. We then generate topic-specific keywords and encode each document's topic membership (and confidence probability) as input features.

Temporal Data Processing: We mine the temporal structure of citation data to capture evolutionary trends. Explicitly, we process the `citations_by_year` attribute into fixed-length time-series sequences. We also compute derivative metrics such as the slope of citation growth for papers, categories, and topics over time. These temporal features enable the capturing of temporal dependencies—crucial for forecasting future impact in our recurrent neural network models.

4 Methodology and Models

In this study, we approach citation prediction as a time-series forecasting task, utilizing sequence models including Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). Our objective is to predict the citation counts of research papers for multiple future years based on their historical data. This approach is selected because citation accumulation is an inherently temporal phenomenon; sequence models are specifically designed to capture the temporal dependencies and evolving trends that static models often miss, thereby providing a robust framework for long-term impact analysis.

4.1 Data and Training Configuration

To ensure a fair and consistent comparison, all sequence models are trained using the same data and training configuration. The models adopt a hybrid input structure, combining dynamic time-series features with static metadata features.

- **Dynamic Features (Time-Series Data):**
 - Citation histories from 2012-2019 are used as sequential inputs to predict citation counts in 2020.
 - Data are reshaped into a 3D tensor with dimensions (samples, time steps, features).
 - The number of time steps is 8, corresponding to eight years of citation history.
 - The feature dimension is 1 (citation count per year).
 - A Masking layer with a mask value of -1.0 is applied to handle missing temporal data for papers not published in certain years.



- **Static Features:**

- Non-temporal metadata, including topic information, number of pages, and category labels are used as static inputs.
- All static features are standardized using StandardScaler to achieve zero mean and unit variance prior to training.

- **Target Variable:** The prediction target is the number of citations in next k years from 2020 (in our case, $k = 5$, i.e., 2020-2024).

- **Data Split:** Samples from 2012-2019 are split into 80% training and 20% validation sets.

- **Training Configuration:**

- Loss function: Mean Squared Logarithmic Error (MSLE), chosen for robustness to right-skewed citation distributions and reduced sensitivity to outliers.
- Optimizer: Adam with an initial learning rate of 0.001.
- Training setup: Batch size 128, maximum 300 epochs.
- Regularization: Early stopping with 30-epoch patience, monitoring validation loss and restoring best weights.
- Learning rate scheduling: ReduceLROnPlateau with a 0.5 decay factor after 10 epochs without validation improvement; minimum learning rate $1e-5$.

- **Model Architecture:**

- Recurrent Neural Network (RNN): The simplest recurrent architecture, where the hidden state is computed from the current input and the previous hidden state; effective for short-term dependencies but limited in modeling long-range dependencies.
- Long Short-Term Memory (LSTM): An extension of standard RNNs that employs memory cells and gating mechanisms (input, forget, and output gates) to preserve and control information over long sequences, enabling better modeling of long-range dependencies.
- Gated Recurrent Unit (GRU): A streamlined alternative to LSTM that combines input and forget gates into a single update gate and uses a reset gate to regulate information flow, reducing model complexity while maintaining temporal modeling capability.

4.2 Evaluation

We evaluate the models on the following metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2). The results are shown in Table 3. Other observations about model predictions, contribution into Sum of Squared Error (SSE) and feature attribution are presented in Figure 4a, 4b and 4c. Some key observations from the results:

- **Model Performance:** LSTM and GRU achieved the best performance across all years on the test set. This is expected since LSTM and GRU are more advanced architectures that can better capture temporal dependencies and patterns in the data.
- **Error Accumulation:** Prediction error increased annually due to the recursive nature of the forecasting (where errors in year t propagate to year $t+1$).



Year	RNN				LSTM				GRU			
	MSE	RMSE	MAE	R ²	MSE	RMSE	MAE	R ²	MSE	RMSE	MAE	R ²
43.64 2020	739.40	27.19	2.60	0.95	43.64	15.54	2.18	0.98	387.84	19.69	2.33	0.97
2021	14753.69	121.46	4.46	0.93	1864.66	43.18	3.22	0.99	463.49	21.53	3.10	1.00
2022	45388.61	213.05	6.46	0.81	6553.61	80.95	4.51	0.97	33510.36	183.06	6.09	0.86
2023	96553.16	310.73	8.33	0.62	6480.56	80.50	5.49	0.97	48976.54	221.31	7.26	0.81
2024	133359.03	365.18	9.20	0.47	61885.76	248.77	7.78	0.75	43143.24	207.71	7.61	0.83

Table 3: Evaluation of models on test set

- Bias: The model exhibits a downward bias toward zero (see Figure 4a), which is likely due to the heavily right-skewed dataset (as explicitly shown in Figure 3) where the majority of papers have very low citations.
- Error Concentration: The total Sum of Squared Errors (SSE) is 237934.93, with the top 1% of papers accounting for 215098.24 (90.40%) of the total error, as shown in Figure 4b. This indicates that the model’s performance metrics are disproportionately influenced by a small number of high-citation outliers.
- Feature Attribution: The SHAP (SHapley Additive exPlanations) values illustrated in Figure 4c offer a granular view of feature importance within the LSTM architecture. The results in Figure 4 indicate that historical citation counts (such as `citations_2019` and `citations_2018`) and temporal context (`num_years_after_publication`) emerge as the most influential variables. In contrast, the semantic features extracted from the paper’s title and abstract (e.g., `title_43`, `abstract_77`) exhibit substantially lower importance scores. This finding implies that the model behaves primarily as a time-series extrapolate, relying heavily on the past citations rather than the intrinsic semantic content or specific research topics of the papers.

5 Conclusion

Our approach of using sequence models provide a stable baseline for citation forecasting but faces significant challenges with “high-impact” papers that experience sudden citation spikes. Ultimately, the models rely on metadata (authors, past citations, venue ranking) rather than the intrinsic quality of the research content. Therefore, these predictions should be viewed as a statistical reference rather than a definitive forecast of a paper’s future impact.

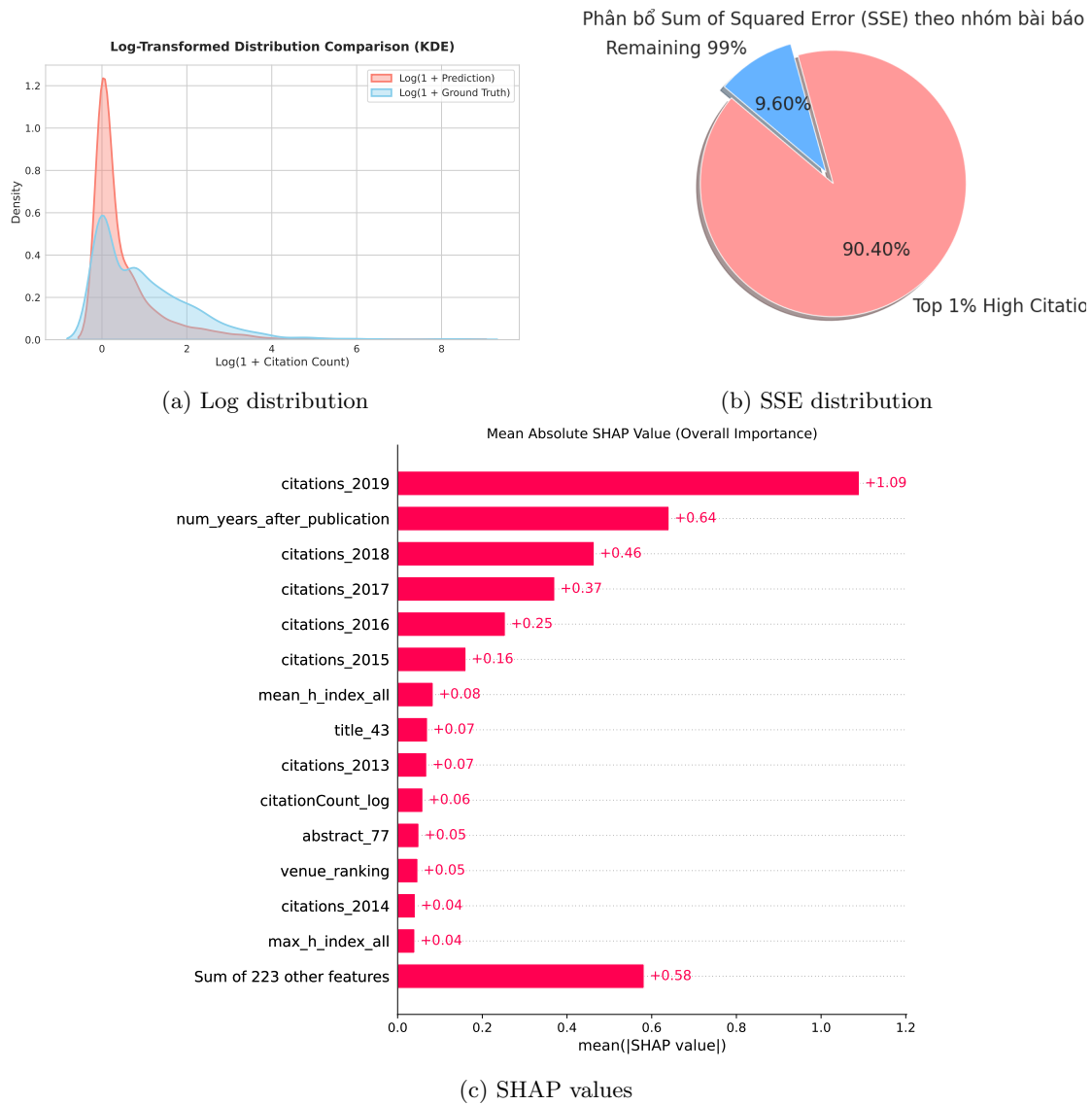


Figure 4: Model prediction analysis: log distribution of citation counts, SSE distribution, and SHAP values for RNN model.